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Spacecraft Attitude Dynamics Individual Assignment

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Abstract

The purpose of this report is the first approximate design of an attitude control system of a spacecraft. The work has been done by taking as a reference one of the satellites belonging to the ESA's GALILEO constellation, the European designed Global Navigation Satellite System. The main objective will be to de-tumble the spacecraft, manoeuvre it and make it point the Earth, as it does in the real life. Below are the specifications and the adjustments that have been done:

Table 1: Project Specifications

	Specification	**Modifications**	**Motivation**
Attitude Parameters	Quaternions	-	-
Mandatory Sensors	Sun Sensor	Earth Sensor and Gyroscopes	Full attitude reconstruction
Actuators	Variable Thrusters	-	-

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Introduction

1.1 Operational Context

Satellites belonging to the Galileo constellation operate in a MEO environment; they aim at providing an accurate geo-localization with a high level of reliability and a good coverage all around the globe. For this reason they are quite large platforms with the need to accomodate powerful instruments. The constellation consists in satellites placed in circular orbits on three different planes, all inclined with respect to the Earth's Equator. The specific orbit taken as reference is listed in Table 1.1. These satellites rely on a three-axis stabilized attitude control system, which is required since they shall be able to perform the following tasks:

- **De-tumbling**: the satellite shall be capable to stop its rotational motion after the detachment from the launcher.
- **Slew manoeuvre**: the satellite shall be capable to point a desired target starting from a condition of null rotational motion.
- **Tracking**: the satellite shall be capable to maintain the pointing of the target.

Table 1.1: MEO orbit keplerian elements

a [km]	**e** [-]	**i** [°]	$\mathbf{\Omega}$ [°]	$\boldsymbol{\omega}$ [°]	$\boldsymbol{\theta}$ [°]
29600	0	56	125	0	0

1.2 Satellite Geometry and Mass Properties

Galielo satellites consists in a main body shaped as a parallelepiped and two solar appendages that produce the required electric power. The platform isn't cuboidal, but its three dimensions are all different from each other. Approximate geometrical values are available to the public, and are reported in Table Table 1.2, where the solar arrays have been assumed as 2-D objects. Their actual mass has been estimated from their area by assuming a density of 2 [kg/m^2], an indicative value based on average densities declared on the web by solar panels manufacturers, such Spectrolab, declare.

The total inertia J of the satellite in the principal axis frame is computed by summing the contributions of the main body and of the solar arrays as reported below in Eq.

Eq. (1.1), and it is assumed to remain constant throughout the simulation. The center of mass similarly has been considered fixed: propellant consumption have not been accounted for. Eq. Eq. (1.2) reports the computation of the mass moments of inertia given the mass, assumed uniformly distributed, and the geometry.

Figure 1.1: Satellite of the Galileo constellation

Table 1.2: Satellite characteristics

m_{mb} [kg]	m_{sa} [kg]	w_{mb} [m]	h_{mb} [m]	l_{mb} [m]	w_{sa} [m]	h_{sa} [m]	l_{sa} [m]
700	27	1.2	1.1	2.7	4.5	1.5	0

$$J = \begin{pmatrix} I_{xx,mb} & 0 & 0 \\ 0 & I_{yy,mb} & 0 \\ 0 & 0 & I_{zz,mb} \end{pmatrix} + \begin{pmatrix} I_{xx,sa} & 0 & 0 \\ 0 & I_{yy,sa} & 0 \\ 0 & 0 & I_{zz,sa} \end{pmatrix} \tag{1.1}$$

$$\begin{gathered} I_{xx,mb} = \frac{m_{mb}}{12}(w_{mb}^2 + h_{mb}^2) \\ I_{yy,mb} = \frac{m_{mb}}{12}(h_{mb}^2 + d_{mb}^2) \\ I_{zz,mb} = \frac{m_{mb}}{12}(w_{mb}^2 + d_{mb}^2) \\ I_{xx,sa} = \frac{m_{sa}}{12}w_{sa}^2 + \frac{m_{sa}}{12}(\frac{w_{sa}}{2} + \frac{w_{mb}}{2})^2 \\ I_{yy,sa} = \frac{m_{sa}}{12}h_{sa}^2 \\ I_{zz,sa} = \frac{m_{sa}}{12}w_{sa}^2 + \frac{m_{sa}}{12}(\frac{w_{sa}}{2} + \frac{w_{mb}}{2})^2 \end{gathered} \tag{1.2}$$

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Model Implementation

2.1 Dynamics

The rotational dynamics of the satellite is solved by integrating the Euler's equations for a rigid body, reported in Eq. (2.1),

$$J\frac{d\boldsymbol{\omega}(t)}{dt} = -\boldsymbol{\omega}(t) \times J\boldsymbol{\omega}(t) + \mathbf{T}(t) \tag{2.1}$$

Where here vector $\mathbf{T}$ includes all possible external torques, either disturbances or control actions.

2.2 Kinematics

Quaternions are the attitude parameter used to represent the spacecraft orientation with respect to the inertial frame. Eq. (2.2) shows how their rate of change is computed given the angular velocity vector, obtained from the dynamics equation. The convention adopted uses q_4 as the quaternion scalar part, being $\mathbf{q}$ the vector part made by the first three components. The initial condition for the quaternions integration has been taken as (0; 0; 0; 1).

$$\dot{\boldsymbol{q}} = \frac{1}{2}\Omega\boldsymbol{q}(t)\ ; \quad \begin{bmatrix}\dot{q_1}\\ \dot{q_2}\\ \dot{q_3}\\ \dot{q_4}\end{bmatrix} = \frac{1}{2}\begin{pmatrix}0 & \omega_3 & -\omega_2 & \omega_1\\ -\omega_3 & 0 & \omega_1 & \omega_2\\ \omega_2 & -\omega_1 & 0 & \omega_3\\ -\omega_1 & -\omega_2 & -\omega_3 & 0\end{pmatrix}\begin{bmatrix}q_1\\ q_2\\ q_3\\ q_4\end{bmatrix}\ ; \tag{2.2}$$

2.3 Quaternions and Direction Cosines Matrices

Eq. (2.3) shows the relation between the quaternion obtained from the kinematics and the corresponding attitude matrix. $[q^\wedge]$ the skew symmetric matrix whose elements are from $\mathbf{q}$.

$$A_{B/N} = (q_4^2 - \boldsymbol{q}^{\boldsymbol{T}}\boldsymbol{q})I + 2\boldsymbol{q}\boldsymbol{q}^{\boldsymbol{T}} - 2q_4[q]^\wedge; \tag{2.3}$$

2.4 Simulink Model

Fig. 2.1 represents the model implemented in Simulink. In the environment area we find both the orbit propagation block which contains information regarding the orbit evolution, considered here as a pure 2B orbit, and the perturbations block contains the different sources of disturbance actions. In the other blocks, for example the controller or in the actuator ones we find logics of action that will be described in the next chapters.

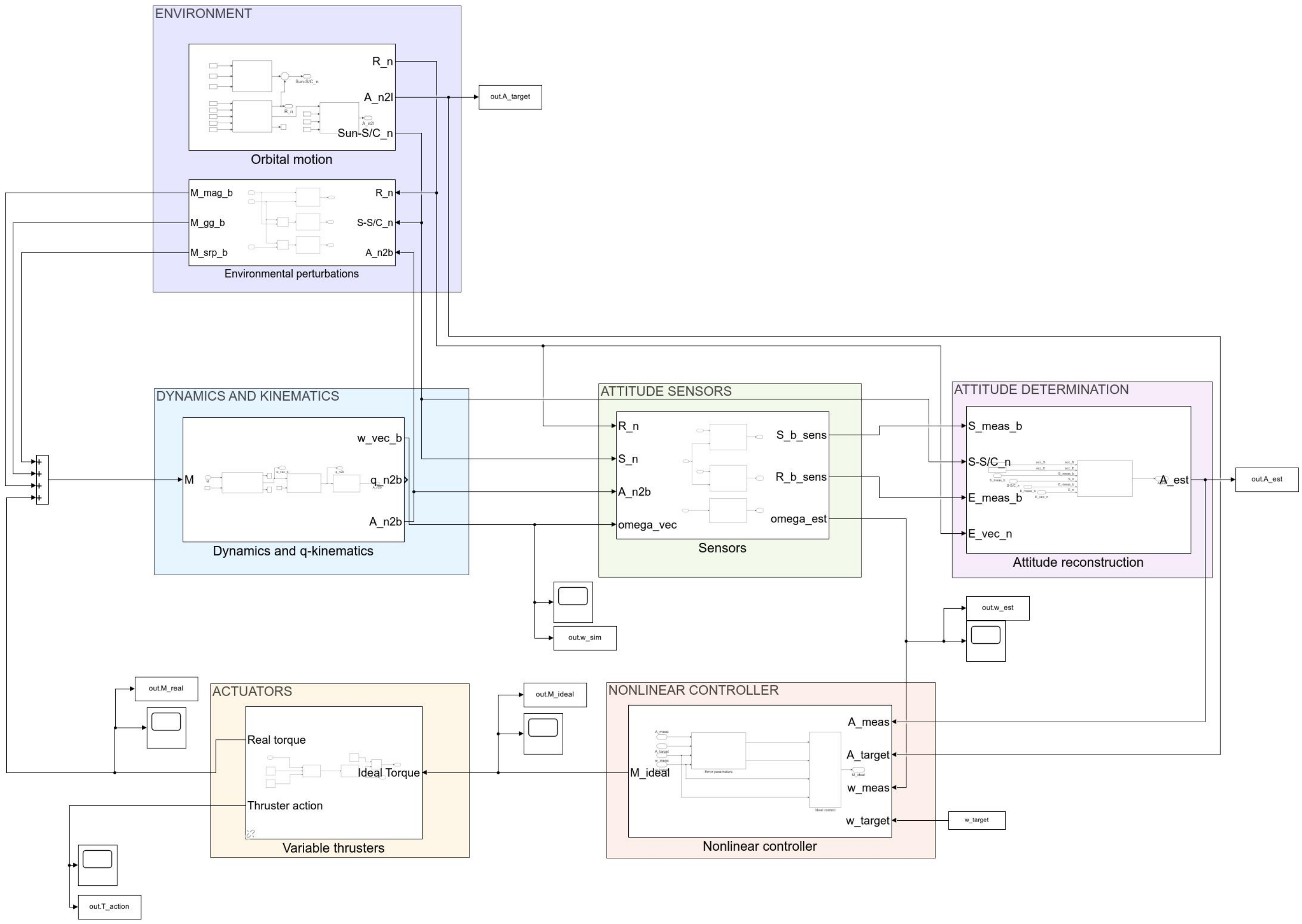

Figure 2.1: Simulink model divided in subsystems

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Environment Perturbations

Given the relatively high altitude of the orbit, the air drag has not been considered for what concerns disturbance torques. The magnetic torque as well is much less important than at lower altitudes but still it has been simulated, at least in a first instance. The next sections will briefly describe the three sources of disturbances acting on the satellite's attitude dynamics.

3.1 Gravity Gradient

Earth's gravity field does not affect uniformly the entire satellite, owing to the non-symmetrical distribution of mass, generating a torque where the symmetry axis is aligned with the gravity vector. Eq. (3.1) represents an approximated expression for this peculiar disturbance, where r is the spacecraft distance from the Earth center.

$$T_{GG} = \frac{3\mu}{r^3}\hat{r} \times J\hat{r}; \tag{3.1}$$

3.2 Solar Radiation Pressure

The SRP perturbation depends on the amount of solar radiation impacting on the spacecraft surfaces. In this scenario only the direct solar radiation has been considered. The spacecraft has been discretized using flat surfaces, for each of which the associated generated torque is computed provided it is actually facing the Sun direction. No self-shadowing was taken into account. ρ_S and ρ_D are respectively the specular and diffuse reflectivities, depending on the surface materials, while P is the average radiation pressure. A slight shift between the geometrical center and the center of mass has been introduced only in this specific case, otherwise the satellite's symmetry would have made the resulting torque null. Eq. (3.2) represents the expression of the force generated by the SRP on an enlightened surface.

$$F_i = -PA_i\hat{s}_b \cdot \hat{n}_i\Big((1-\rho_s)\hat{s}_b + 2\Big(\rho_s\hat{s}_b \cdot \hat{n}_i + \frac{1}{3}\rho_D\Big)\hat{n}_i\Big); \quad T_{SRP} = \sum_{i=1}^{n_{surf}} F_i \times d_{CM,i}; \tag{3.2}$$

3.3 Magnetic Perturbation

The magnetic disturbance torque is caused by the interaction between Earth's magnetic field, and the residual magnetic induction due to on-board currents through wirings/electronics. The Earth's Magnetic Field has been described by the magnetic dipole approximation, stated in Eq. (3.3), given the altitude of the orbit. The parasitic induction is difficult to model, and has been taken as $\hat{m} = [0.5; 0.5; 0.5]\ Am^2$.

$$\mathbf{B} = \frac{R_\oplus H_0}{r^3}(3(\hat{m}\cdot\hat{r})\boldsymbol{r} - \hat{m}); \quad T_{mag} = \mathbf{m}\times\mathbf{B}; \tag{3.3}$$

Table 3.1: Magnetic Model Constants

	Value	Symbol
Magnetic Parameter	3.011×10^{-5}	H_0
Earth Magnetic Dipole	$\begin{bmatrix} sin(11.5°)cos\Omega_\oplus t \\ sin(11.5°)sin\Omega_\oplus t \\ cos(11.5°) \end{bmatrix}$	$\hat{m}$
Earth Angular Velocity	$15°$/h	$\Omega_\oplus$

3.4 Disturbance Torques Magnitude

A simulation of one complete orbital period without any active control has been performed in order to get an idea of the orders of magnitude of the perturbations discussed so far. Fig. 3.1 and Fig. 3.2 show the results: the magnetic torque is negligible with respect to the other ones. Still all of them would need a good amount of time to produce noticeable effects on the attitude.

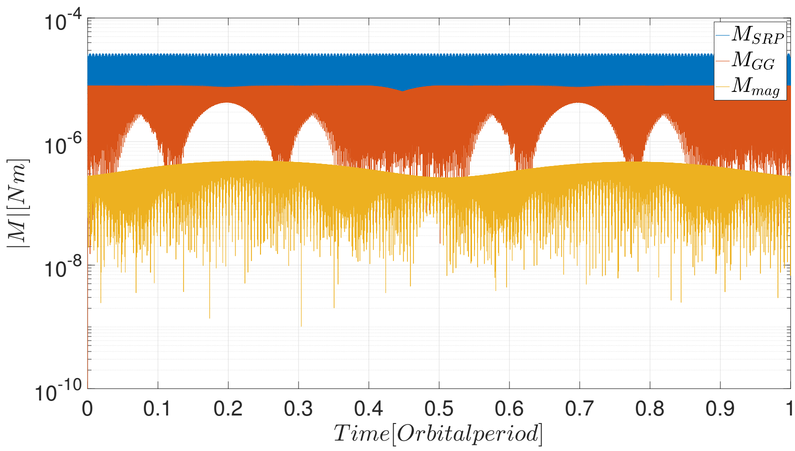

Figure 3.1: Magnitude of disturbance torques

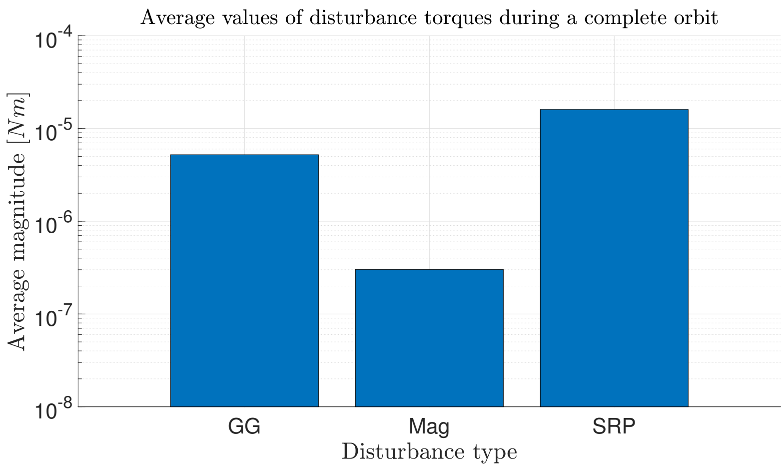

Figure 3.2: Average disturbances over one orbit period

4

Sensors

A Sun Sensor alone would not allow to fully reconstruct the attitude of the spacecraft. For this reason an Earth-Horizon Sensor has been added. Moreover, since the efforts to reconstruct the angular velocity from their measurements using some numerical estimates did not deliver results accurate enough, also a set of gyroscopes has been added to the system.

4.1 Sun Sensor & Hearth-Horizon Sensor

Due to their complex working principles, the internal dynamics has not been simulated. Their measures instead have been considered to be the "true", or better the simulated ones, rotated by a matrix A_ε depending on three random angles. These are generated from a Gaussian distribution and the accuracy reported on data sheets refers to its 1σ amplitude. Table 4.1 and Table 4.2 provide examples of two off-the-shelf models of these sensors.

Table 4.1: Sun Sensor - SS200 (*Hyperion Technologies, B.V.*)

Accuracy	**FoV**	**Sampling rate**	**Power (@idle)**	**Power (@sampling)**	**Mass**	**Dimensions**
$< 1°$	110°	100 Hz	< 1.5 mW	$2.5 \div 40$ mW	3 g	$(24.7 \times 15.0 \times 3.5)$ mm

Table 4.2: Earth Horizon Sensor - MAI-SES (*Maryland Aerospace, Inc.*)

Configuration	**Accuracy**	**FoV**	**Voltage**	**Current (per sensor)**	**Mass**	**Dimensions**
4 thermopile detectors	$< 0.25°$	7°	3.3 V	40 mA	33 g	$(43.3 \times 31.7 \times 31.7)$ mm

4.2 Gyroscopes

In order to get measurements on the angular rate, three gyroscopes have been places along the satellite axes assuming no misalignment. Thir output is usually affected by two sources of errors as shown in Eq. (4.1).

$$\dot{\omega}_{meas} = \omega_{sim} + n + b \tag{4.1}$$

Here, ω is the angular velocity of the spacecraft coming from the dynamics. Additionally, n and b represent two *white noise terms*, which are respectively the Angular Random Walk (ARW) and Rate Random Walk (RRW). Their is computed as defined in Eq. (4.2).

$$\sigma_n = \frac{ARW}{\sqrt{1/f}}; \quad \sigma_b = \frac{RRW}{\sqrt{1/f}} \tag{4.2}$$

Where f is the sampling frequency. Table 4.3 lists these parameters for an example of gyroscope available on the market.

Table 4.3: Gyroscope Specifications

Model name	**ARW** $[°/\sqrt{hr}]$	**RRW** $[°/hr^{3/2}]$
ASTRIX 200	0.0001	0.00016

In the computational model all these sensors' frenquencies have been set to $10[Hz]$ for numerical reasons: a higher sampling rate would have made the computational burden to skyrocket.

4.3 Attitude Determination Algorithm

The SVD method has been chosen as the algorithm that reconstructs the attitude in a statistical sense, meaning that takes into account the relative precision of the various sensors. Such algorithm provides a solution to the Wahba's problem, or the minimization of the following cost function:

$$\mathcal{J} = \frac{1}{2}\sum_{i=1}^{N} \alpha_i \|\hat{\underline{v}}_{B_i} - A_{B/N}\hat{\underline{v}}_{N_i}\|$$

In which N sensor measurements $\hat{\underline{v}}_{B_i}$ are available, along with the corresponding estimations $A_{B/N}\hat{\underline{v}}_{N_i}$ computed using on-board models; in addition the weighting coefficients α_i are chosen based on the accuracy of each sensor.

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Control

5.1 Lyapunov Controller

A general, non-linear function for attitude control is the Lyapunov controller, which is based on minimising a scalar function V of the attitude state, that reaches 0 value when the desired attitude is achieved. This control law holds for all the three mission phases: there only the need to change two gains at a time. But first, we need to retrieve the error quantities that express the difference between the desired target or angular velocity and the measured one

$$A_{err} = A_{meas}A_{req}^T \ ; \quad \omega_{err} = \omega_{meas} - A_{err}\omega_{req} \ ; \tag{5.1}$$

The minimization function used is:

$$V = \frac{1}{2}k_1\omega_{err}^T J\omega_{err} + 2k_2 tr(I - A_{err}) \ ; \tag{5.2}$$

Deriving Eq. (5.2), the ideal control torque is obtained:

$$T_{id} = -k_1\omega_{err} - k_2(A_{err}^T - A_{err})^V - J\omega_{meas} \times \omega_{meas} + J(A_{err}\dot{\omega}_{req} - [\omega_{err}]^\wedge A_{err}\omega_{req}) \ ; \tag{5.3}$$

The parameter k_1 is proportional to the angular velocity error, while k_2 relates to the attitude position error. In this way, it is possible to control the satellite by changing them accordingly to the desired attitude mode.

Table 5.1: Controller weights

Mode	$\mathbf{k}_1$	$\mathbf{k}_2$
De-tumbling	6	0
Slew manoeuvre	5.6	0.1
Tracking	5.6	0.1

The disadvantage of this method is that a proper selection of these gains is not trivial. Furthermore despite proving the system stability, no real performance parameter comes easily from them. As a matter of fact the values reported in Table 5.1 have been chosen with a trivial trial and error procedure.

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Actuators

6.1 Thrusters

An essential part of the design of a thruster control system lies in the definition of the geometrical configuration. The relation between each thruster and the global resulting torque can be expressed in general as in Eq. (6.1).

$$T_{out,T} = \hat{R} F_{out,T}; \tag{6.1}$$

$F_{out,T}$ is the force produced by the thrusters at each time step and can't be negative. An equation that retrieves the thrusters action making sure of their sign involves the pseudo-inverse matrix of $\hat{R}$:

$$F_{out,T} = \hat{R}^* \, T_{ideal} + \gamma\omega; \tag{6.2}$$

Where $\gamma\omega$ is a null space vector (made of unitary elements) used to avoid singularities when inverting the matrix $\hat{R}$. A simple 4 thruster configuration has been selected. Despite not having a null net thrust, usually such effect is only a low-order perturbation of the orbital motion, and thus it has not been taken into account in our analysis. Its $\hat{R}$ is reported in Eq. (6.3).

$$\hat{R} = \begin{pmatrix} lsin\alpha & 0 & 0 \\ 0 & lcos\alpha & 0 \\ 0 & 0 & xsin\alpha - xcos\alpha \end{pmatrix} \begin{pmatrix} 1 & 1 & -1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix}; \tag{6.3}$$

An upper bound on the thrust of $F_{max} = 10\ N$ has been selected, with a lower bound of $F_{min} = 0.05\ N$. Additionally, a zero-order hold block have been used to simulate a MIB value estimated as $0.03\ s$. In this way the real torque produced by the actuator is not guaranteed to be exactly the ideal one decided by the attitude controller.

7

Results

7.1 De-tumbling

During de-tumbling the satellite has to lose all the residual angular velocity acquired during the in-orbit release. Therefore, the target angular velocity is zero, while no conditions are imposed on the reference attitude. To do so, the S/C controller starts *mode 1* at 0s. The initial angular velocity has been set to $\omega = [0.3, 0.3, 0.3]^T$, which comes from the Ariane 5 user manual. Fig. 7.1 and Fig. 7.2 show the evolution of the angular velocity and the thrust required from each engine for this mission phase. The satellite is able to reach a low angular speed in a reasonable time.

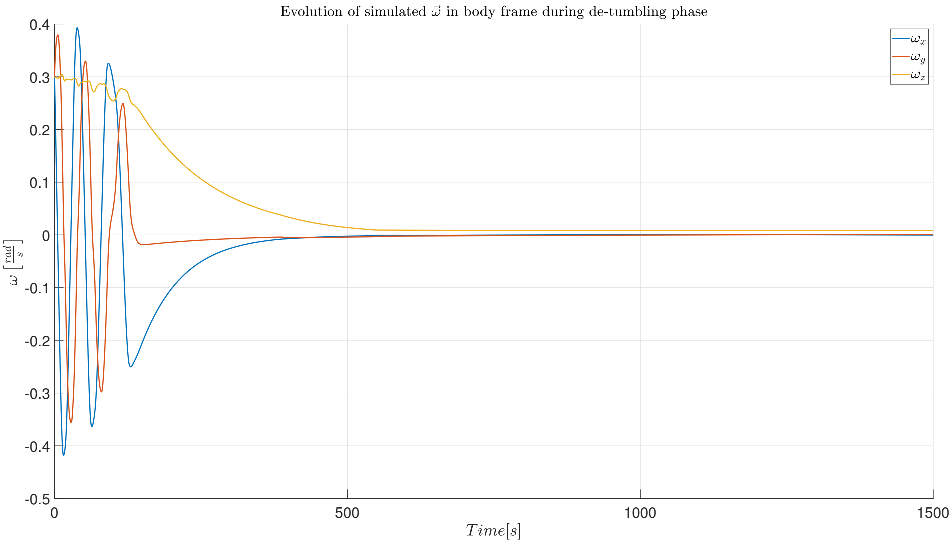

Figure 7.1: Angular velocities during de-tumbling phase

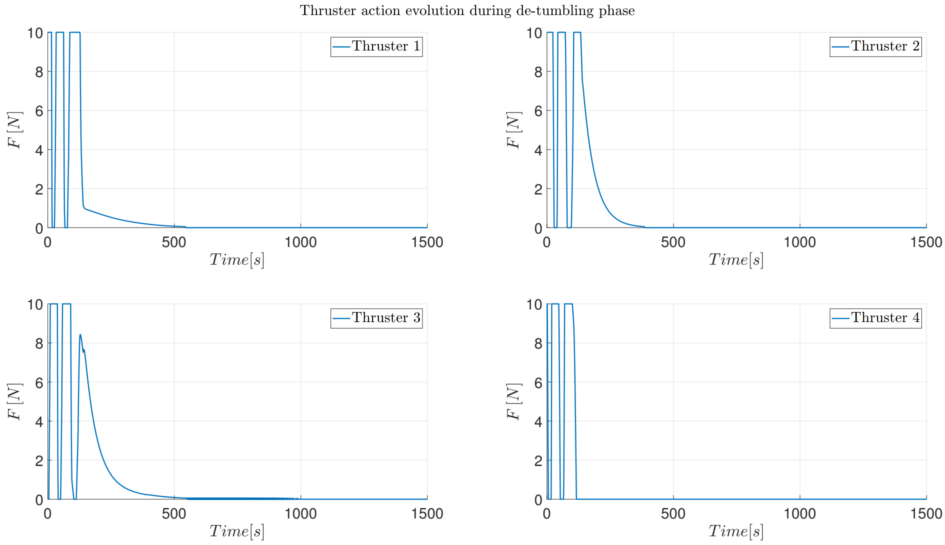

Figure 7.2: Angular velocities during de-tumbling phase

7.2 Slew Manoeuvre

Aim of the slew phase is to bring the satellite from one attitude configuration to another. Specifically, from the one at end of the de-tumbling phase to the alignment with respect to the LVLH frame. It is required that the angular velocities coincide as well. For this reason in this phase the Lyapounov controller sets both its gains to values different from zero. As seen in Fig. 7.3 and Fig. 7.5 such goal can be completed and the system remains quite stable.

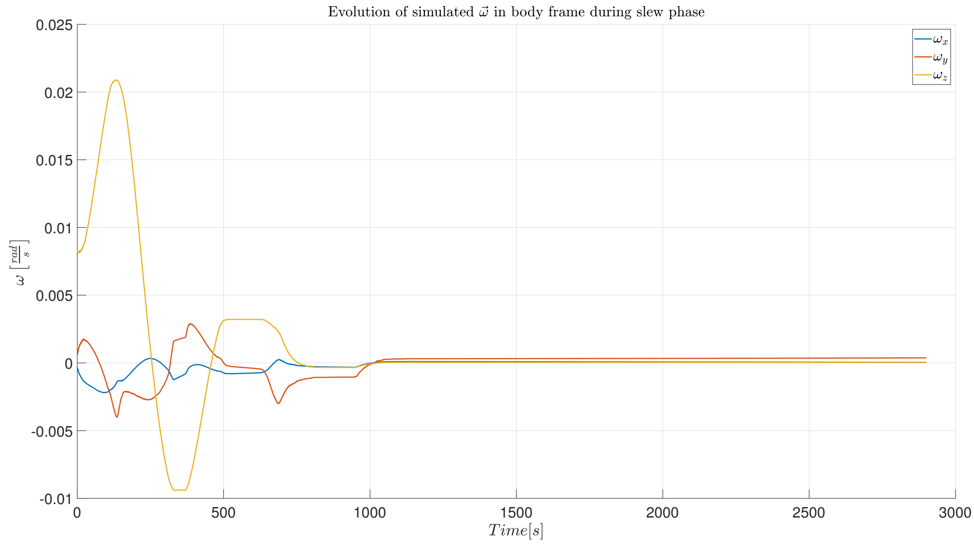

Figure 7.3: Angular velocities during sl

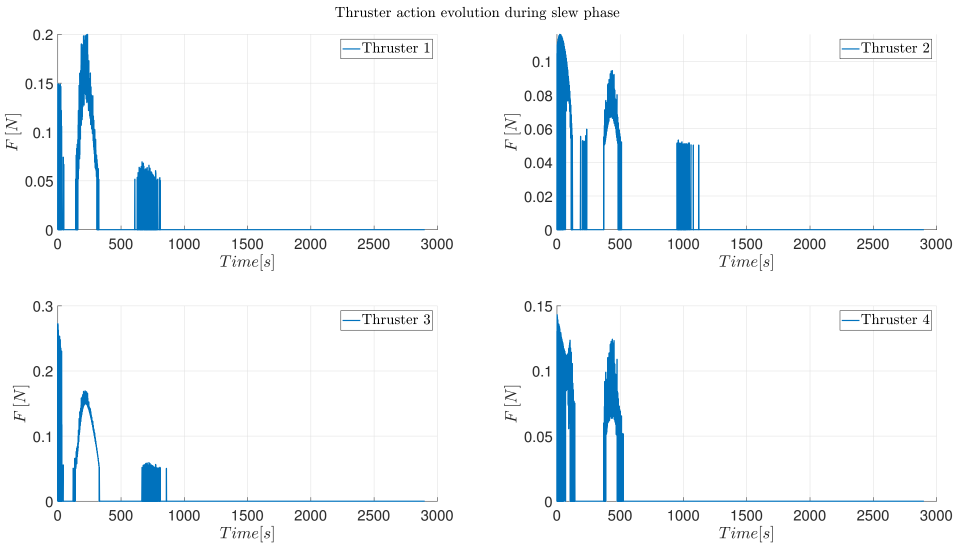

Figure 7.4: Angular velocities during slew manoeuvre

7.3 Tracking

This phase starts the operative life of the satellite. After it is completed, the S/C is correctly positioned towards the Earth, and it can communicate with on-ground devices. The requirement on satellite attitude is for the antenna to be nadir pointing. The satellite angular velocity thus should match the orbital one.

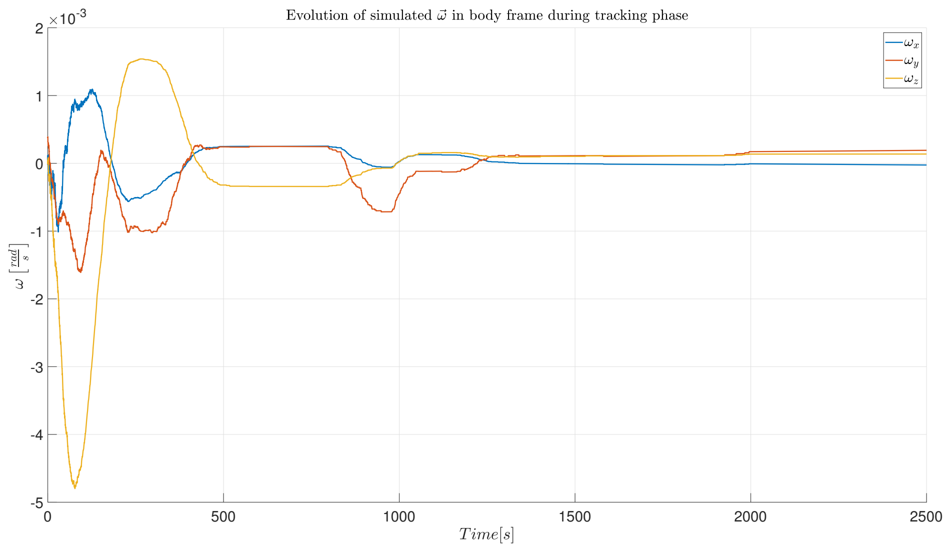

Figure 7.5: Angular velocities during tracking

8

Conclusions

The mission objectives have been successfully accomplished with the implementation of the aforementioned manoeuvres of de-tumbling, slew, and reference tracking. The results have shown that the attitude errors converge in a fraction of the orbital period. However, the high value of the minimum thrust that each engine can provide is a limiting factor on the maximum precision that the system can achieve. Adopting finer thrusters capable of delivering lower control torques would increase the global accuracy of the results, in particular for what concerns the tracking phase. Moreover, studies can be made to account for perturbations affecting orbital mechanics; the number of thrusters can be increased in order to provide null net thrust on the satellite; the center of mass can be assumed to have an offset without loss of generality, making it possible to have a decrease of the mass, therefore inertia, over the lifetime of the spacecraft.